THE MUTATION GRAPHS OF UNIFORM ORIENTED MATROIDS ON AT MOST NINE ELEMENTS ARE CONNECTED

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Abstract. Cordovil and Las Vergnas conjectured that the mutation graph of uniform oriented matroids of fixed rank and size is connected. Knauer and Marc distinguish three versions of this graph — on labelled uniform oriented matroids, on their reorientation classes, and on their isomorphism classes — prove the labelled graph connected in rank 3 for every n, prove the conjecture for $n \leq 9$ at the reorientation-class level, and write that beyond rank 3 they know nothing about the labelled graph, where they suspect a counterexample exists. We show there is no counterexample below $n = 10$: for every rank r and every $n \leq 9$, all three graphs are connected. Once their rank-3 theorem, oriented-matroid duality and the elementary ranks ($r \leq 2$ and $r \geq n-1$) are accounted for, the cases genuinely settled here are $(n, r) = (8, 4)$, $(9, 4)$ and its dual $(9, 5)$ — among them the case they single out as hardest. The method is a covering-space reduction expressing the number of components of the labelled graph as the index of a holonomy subgroup computable on the isomorphism-class quotient. Two by-products: the number of isomorphism classes of uniform rank-4 oriented matroids on 9 elements is 9,276,595, proved by a mass identity against a structurally independent extension count, correcting the value 9,276,601 printed by Fukuda, Miyata and Moriyama; and, when the isomorphism-class graph is connected and the holonomy surjects onto S_n, for odd n the number of components of the labelled graph is forced into $\{1, 2^{n-1}\}$ with no intermediate value possible. All computations are exact. Certificates and standalone checkers, all of them in the repository, accompany the holonomy claims, the 9,276,595-class enumeration, and a mutation spanning tree joining those classes into a single component; §6 states precisely which claims they cover and which remain reproducible rather than certified.

1. Introduction

Let $\mathcal{M}$ be a uniform oriented matroid (UOM) of rank r on $E = \{1, \ldots, n\}$, presented by a chirotope χ. A *mutation* of $\mathcal{M}$ flips the sign of a single basis so that the result is again a chirotope; geometrically, for realizable $\mathcal{M}$, it is the passage of one point through the hyperplane spanned by $r-1$ others. The *mutation graph* has the UOMs of rank r on n elements as vertices and mutations as edges.

Cordovil and Las Vergnas conjectured that this graph is connected; the conjecture is recorded in [4]. Knauer and Marc [3] observe that the statement depends on which objects one takes as vertices, and separate three graphs, in decreasing fineness:

$\overline{\mathcal{G}}^{n,r}$ (labelled UOMs), $\qquad \mathcal{G}^{n,r}$ (reorientation classes), $\qquad \underline{\mathcal{G}}^{n,r}$ (isomorphism classes),

where reorientation means the action of $\{\pm 1\}^n$ (sign changes) and isomorphism the action of the full hyperoctahedral group $\{\pm 1\}^n \rtimes S_n$ (sign changes and relabelling). Connectivity of each implies connectivity of the next. Knauer and Marc state the Cordovil–Las Vergnas conjecture at the middle level. They prove, at the *labelled* level, that $\overline{\mathcal{G}}^{n,3}$ is connected for every n [3, Prop. 3.2], via Ringel's homotopy theorem together with an explicit reorientation gadget; and they deduce $\mathcal{G}^{n,r}$ connected for $n \leq 9$ from a computation that $\underline{\mathcal{G}}^{n,r}$ is connected

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in that range. Of the finest level in higher rank they write that checking connectivity "is far more demanding" and that they "do not know anything about the connectivity of this graph beyond rank 3"; in their conclusion they say they "suspect the existence of a counter example at least in the setting of $\overline{\mathcal{G}}^{n,r}$."

Our main result is that no such counterexample exists below $n = 10$.

Theorem 1. *For every $n \leq 9$ and every rank r, the graphs $\overline{\mathcal{G}}^{n,r}$, $\mathcal{G}^{n,r}$ and $\underline{\mathcal{G}}^{n,r}$ are connected.*

The content is the labelled level; the other two follow. Much of the range is not new, and it is worth saying at once which part is. Ranks $r \leq 1$ and $r \geq n-1$ are degenerate and elementary (§5); rank 2 is elementary as well; rank 3 is [3, Prop. 3.2], and rank $n-3$ follows from it by oriented-matroid duality. What remains at $n \leq 9$ is rank 4 for $n = 8, 9$ together with $(9, 5)$, which is dual to $(9, 4)$. So the cases genuinely settled here are

$$(n, r) \ = \ (8, 4), \qquad (9, 4), \qquad (9, 5) \cong (9, 4)^*,$$

and it is $(9, 4)$ that carries the weight: its isomorphism-class quotient has 9,276,595 vertices, and it is the case Knauer and Marc identify as beyond reach. Theorem 1 does not settle the Cordovil–Las Vergnas conjecture, nor the labelled question, in general: it rules out a counterexample only for $n \leq 9$.

Two by-products deserve separate statement. The first corrects the literature.

Theorem 2. *The number of isomorphism classes of uniform oriented matroids of rank* 4 *on* 9 *elements is exactly* 9,276,595.

This value appears in Finschi's database [1] and in [3, Table 1]; Fukuda, Miyata and Moriyama [2, Table] print 9,276,601, which is six too many. Our argument is not a third opinion but an exact integer identity: the orbit masses of the classes found by an exhaustive search sum to a target obtained by a structurally independent route, a single-element extension sweep over the 2,628 classes at $n = 8$. Under hypotheses made explicit in §5 — the enumerated classes valid, pairwise inequivalent, and carrying their true stabilizer orders, and the target exact — a search that missed a class falls short and cannot overshoot. The first three hypotheses are discharged by a standalone checker over all 9,276,595 classes; the fourth is reproducible but not independently certified (§6).

The second by-product explains why the labelled question at odd n is binary.

Proposition 3. *Let n be odd, let $\underline{\mathcal{G}}^{n,r}$ be connected, and suppose the holonomy subgroup H of §3 surjects onto S_n. Then*

$$\#\text{components}\big(\overline{\mathcal{G}}^{n,r}\big) \in \{1,\ 2^{n-1}\}.$$

For $n = 9$: either 1 *or* 256*, with no intermediate value possible.*

Finally, we record an erratum: [3, Table 1] lists 482 classes at $(n, r) = (9, 3)$. The correct value is 4382, as its own dual entry at $(9, 6)$, Finschi's database, [2] and our independent generation all agree.

Verifying this note before reading it. The computational claims are machine-checkable, within the scope §6 sets out. From the repository

`https://github.com/reuellee/finite-certificates`

(directory `ai/omgamma`), with Python 3 and no third-party packages beyond `numpy` for the fast checker:

```
python verify_omgamma.py             # holonomy certificates
python verify_omgamma.py --canary    # self-test: must fail on a
                                     # missing artifact
python submodules.py                 # Proposition 3, two methods
python coverage_checker.py --artifact data/coverage_4_9 --workers 2 \
       --extcount data/extcount_4_9.jsonl  # coverage + reachability,
                                           # all 9.28M classes
python coverage_checker.py --canary --artifact data/coverage_4_9
                                           # 12 sabotages; must REJECT
```

Every artifact these commands read is in the repository, the coverage certificate included: it is a 10.4 MB file from which the checker reconstructs all 9,276,595 class keys and stabilizer orders. §6 says how far each command reaches. The ledger `ai/omgamma/OMGAMMA.md` records every computation, its artifacts, its trust boundaries, and its errata — including one of our own, discussed in §6.

2. The three graphs

Fix n and r. A uniform chirotope is an alternating map $\chi\colon E^r \to \{\pm 1\}$ satisfying the three-term Grassmann–Plücker sign conditions; χ and $-\chi$ describe the same oriented matroid, so a vertex of $\overline{\mathcal{G}}^{n,r}$ is the pair $\{\chi, -\chi\}$. Let

$$G' \;=\; S_n \times \{0,1\}^n \times \{0,1\},$$

$$\big((\sigma,\varepsilon,s)\cdot\chi\big)(x_1,\dots,x_r) \;=\; (-1)^s\,(-1)^{\sum_{i=1}^r \varepsilon_{x_i}}\,\chi(\sigma^{-1}x_1,\dots,\sigma^{-1}x_r),$$

where $\varepsilon = (\varepsilon_1,\dots,\varepsilon_n) \in \{0,1\}^n$ records which ground-set elements are reoriented and s is the global sign. Here G' carries the semidirect-product multiplication induced by the S_n-action on $\{0,1\}^n$,

$$(\sigma_1,\varepsilon_1,s_1)(\sigma_2,\varepsilon_2,s_2) = (\sigma_1\sigma_2,\ \varepsilon_1 \oplus \sigma_1(\varepsilon_2),\ s_1 \oplus s_2).$$

Throughout Sections 3–5 we work in the nondegenerate range $2 \le r \le n-2$; the degenerate ranks are disposed of directly in Section 5. There the subgroup $K_4 = \{(\mathrm{id},0,0),(\mathrm{id},1^n,0),(\mathrm{id},0,1),(\mathrm{id},1^n,1)\}$ acts trivially on pairs $\{\chi,-\chi\}$, and we write $\overline{G} = G'/K_4$, of order $n!\,2^{n-1}$, with normal sign subgroup $\bar{R} = \{0,1\}^n/\langle 1^n\rangle$ and quotient $\pi\colon \overline{G} \twoheadrightarrow S_n$. (Nothing below requires K_4 to be the *full* kernel; it is, in this range, but we use only that it acts trivially.) Then

$$\mathcal{G}^{n,r} = \overline{\mathcal{G}}^{n,r}/\bar{R}, \qquad \underline{\mathcal{G}}^{n,r} = \overline{\mathcal{G}}^{n,r}/\overline{G},$$

and mutation is $\overline{G}$-equivariant, so these quotients are graphs in the usual way.

3. From the labelled graph to a subgroup

The reduction is standard covering-space bookkeeping, stated for a group action that is *not* free — which is exactly where care is needed.

Lemma 4. *Let a finite group $\overline{G}$ act by automorphisms on a graph $\overline{\mathcal{G}}$ with connected quotient $\underline{\mathcal{G}} = \overline{\mathcal{G}}/\overline{G}$. Then $\overline{G}$ permutes the components of $\overline{\mathcal{G}}$ transitively, every component maps onto $\underline{\mathcal{G}}$, and for a base vertex v_0,*

$$\#\mathrm{components}(\overline{\mathcal{G}}) = [\overline{G} : H], \qquad H := \{g \in \overline{G} \;:\; g\cdot v_0 \in \mathrm{comp}(v_0)\}.$$

For a chirotope χ and a basis B at which χ is mutable, write $\mu_B\chi$ for the chirotope agreeing with χ except that its sign on B is negated. If the class of $\mu_B\chi_c$ is c', the *voltage* of that mutation edge is the element $t \in \overline{G}$ with $\mu_B\chi_c = t \cdot \chi_{c'}$ (as pairs); it is well defined up to $\mathrm{Stab}_{\overline{G}}(\chi_{c'})$, and any choice may be fixed.

Lemma 5. *Fix class representatives χ_c and a spanning tree of $\underline{\mathcal{G}}$ rooted at c_0. Define transports $\tau_c \in \overline{G}$ by $\tau_{c_0} = \mathrm{id}$ and, along each tree edge directed away from the root, $c \to c'$ with voltage t, by $\tau_{c'} = \tau_c\, t$. Then H is generated by*

$$\underbrace{\{\tau_c\, u\, \tau_c^{-1} \;:\; u \in \mathrm{Stab}_{\overline{G}}(\chi_c)\}}_{\text{(i) stabilizers}} \cup \underbrace{\{\tau_c\, t\, \tau_{c'}^{-1} \;:\; \textit{every mutation edge } (c,j,c')\}}_{\text{(ii) voltages}}.$$

Proof. ($\supseteq$) Induction along the tree gives $\tau_c \cdot \chi_c \in \mathrm{comp}(v_0)$ for every c, mutations being $\overline{G}$-equivariant. Then $\tau_c u \tau_c^{-1}$ fixes the component vertex $\tau_c \cdot \chi_c$, and $\tau_c t \tau_{c'}^{-1}$ carries $\tau_{c'} \cdot \chi_{c'}$ to $\tau_c \cdot (\mu_B\chi_c)$, a neighbour of $\tau_c \cdot \chi_c$; an element carrying some component vertex into the component stabilizes it. Note this direction needs only that the tree paths exist, not that the search is complete, so elements harvested from a partial search are genuine members of H.

($\subseteq$) Let $g \in H$, witnessed by a walk $v_0 = x_0 \sim x_1 \sim \cdots \sim x_k = g \cdot v_0$ in $\overline{\mathcal{G}}$. Write $x_i = s_i \cdot \chi_{c_i}$ with $s_0 = \mathrm{id}$. Applying s_i^{-1} to the edge $x_i \sim x_{i+1}$ shows $s_i^{-1}s_{i+1} \cdot \chi_{c_{i+1}}$ is a mutation of χ_{c_i}, so $s_i^{-1}s_{i+1} \in t_i\, \mathrm{Stab}_{\overline{G}}(\chi_{c_{i+1}})$ for the voltage t_i of some mutation edge $c_i \to c_{i+1}$. Hence $g = s_k h$ with $h \in \mathrm{Stab}_{\overline{G}}(\chi_{c_0})$ and s_k a product of the $s_i^{-1}s_{i+1}$; inserting $\tau_{c_i}^{-1}\tau_{c_i}$ between consecutive factors rewrites that product as a word in the displayed conjugated families, which telescopes because $\tau_{c_0} = \mathrm{id}$. □

Family (i) is not optional: the action of $\overline{G}$ on chirotopes has nontrivial stabilizers, and omitting the stabilizer conjugates computes a proper subgroup of H. We learned this the hard way (§6).

Combining, with $\underline{\mathcal{G}}$ connected: $\overline{\mathcal{G}}^{n,r}$ is connected iff $H = \overline{G}$. For the middle level, the components of $\mathcal{G}^{n,r} = \overline{\mathcal{G}}^{n,r}/\bar{R}$ are the $\bar{R}$-orbits on the components of $\overline{\mathcal{G}}^{n,r}$; since $\bar{R}$ is normal, $\bar{R}H$ is a subgroup, and Lemma 4 gives

$$\#\mathrm{components}(\mathcal{G}^{n,r}) = [\overline{G} : \bar{R}H] = [S_n : \pi(H)].$$

So the labelled question becomes: compute a subgroup of $\overline{G}$ from data harvested on the quotient, and ask whether it is everything.

4. The dichotomy at odd n

Write $S := H \cap \bar{R}$ for the sign part of H. Since $\bar{R}$ is normal and abelian, conjugation gives $(\sigma,\delta,s)(\mathrm{id},\varepsilon,0)(\sigma,\delta,s)^{-1} = (\mathrm{id},\sigma(\varepsilon),0)$, so S is $\pi(H)$-invariant. If $\pi(H) = S_n$ then S is an $\mathbb{F}_2[S_n]$-submodule of $\bar{R}$.

Proof of Proposition 3. For odd n we have $1^n \notin E$, where $E \leq \mathbb{F}_2^n$ is the even-weight subspace, so $\mathbb{F}_2^n = \langle 1^n \rangle \oplus E$ and $\bar{R} \cong E$. We check directly that E is an irreducible $\mathbb{F}_2[S_n]$-module. Let $0 \neq V \leq E$ be invariant and $0 \neq v \in V$. Since v has even weight and n is odd, v has a coordinate i in its support and a coordinate j outside it; then $v + (i\,j) \cdot v = e_i + e_j$ lies in V. The S_n-orbit of $e_i + e_j$ is all weight-two vectors, and these span E, so $V = E$. Hence $S \in \{0, \bar{R}\}$, and by Lemma 4 the number of components is $[\overline{G} : H] \in \{2^{n-1}, 1\}$. □

The conclusion is corroborated computationally, independently of the argument: `submodules.py` enumerates the S_n-invariant subspaces of $\mathbb{F}_2^n$ for $n = 5, \ldots, 10$ as sums of spans of weight classes, finding exactly four for each n; for $n \leq 7$ it also enumerates *all* subspaces through their reduced row echelon forms and filters them, the totals 374/2825/29212 matching the Galois numbers (so that enumeration is provably exhaustive) and agreeing with the first method where both were run. A deliberately non-invariant subspace is rejected by the invariance test. For even n an intermediate value is admissible; the full sign parts observed at $n = 8$ are therefore not forced by algebra.

Proposition 3 is not needed for the final verdict — the certificate exhibits a full sign span directly — but it explains the shape of the computation: at $n = 9$ the tracked sign dimension jumped from its minimum to its maximum with no intermediate value, and the proposition says no intermediate value was ever available.

5. The computation

Ranks $r \leq 1$ and $r \geq n-1$ are degenerate: the three-term Grassmann–Plücker condition set is empty (it is nonempty only when $r \geq 2$ and $n \geq r+2$), so every sign vector on the $\binom{n}{r}$ bases is a chirotope and every single-bit flip is a mutation; $\overline{\mathcal{G}}^{n,r}$ is then a folded cube, visibly connected. Rank 2 is likewise elementary: a uniform oriented matroid of rank 2 on n elements is a cyclic arrangement of n signed points on a circle, so its chirotope is determined by that cyclic order together with the reorientation, a mutation transposes two neighbours, and adjacent transpositions connect all such data. Rank 3 is [3, Prop. 3.2]. Rank 4 is computed exhaustively as described below. Ranks $5 \leq r \leq n-2$ follow by duality. For a basis $B = \{x_1 < \cdots < x_r\}$ with complement $E \setminus B = \{x_{r+1} < \cdots < x_n\}$, the dual chirotope is

$$\chi^*(x_{r+1}, \ldots, x_n) \;=\; \chi(x_1, \ldots, x_r) \cdot \operatorname{sgn}(x_1, \ldots, x_n),$$

sgn being the sign of the permutation $(x_1, \ldots, x_n)$ of E. Then $\chi \mapsto \chi^*$ is a graph isomorphism $\overline{\mathcal{G}}^{n,r} \cong \overline{\mathcal{G}}^{n,n-r}$ (and likewise for $\mathcal{G}$ and $\underline{\mathcal{G}}$), carrying the mutation at B to the mutation at $E \setminus B$: since $\chi^*(E \setminus B) = \pm\chi(B)$ with a sign depending only on B, negating χ at B is the same operation as negating χ^* at $E \setminus B$, and validity is preserved on both sides. (This was also checked computationally, on all 135 classes at $(8,3)$ among others.) This covers every rank at $n \leq 9$.

For rank 4 at $n \leq 8$ (and, as a control, rank 3 at $n \leq 9$, where the answer is already known from [3, Prop. 3.2]) the quotient $\underline{\mathcal{G}}$ is small enough for an exhaustive breadth-first search harvesting both generator families of Lemma 5 at every class. In each case $H = \overline{G}$, so all three graphs are connected; the rank-3 runs agreeing with [3] is one of this project's regression controls.

The case $(9,4)$ needs 9,276,595 classes and 150,561,898 directed edge traversals, run as a disk-based level-synchronous search. Completeness is certified by a *mass identity*. Every class c contributes an orbit of size $|G_9'|/|\operatorname{Stab}(\chi_c)|$ labelled objects, and the total number of labelled uniform chirotopes is computed by a structurally independent counting route — counting single-element extensions of each of the 2,628 classes at $(8,4)$, which touches neither the mutation graph nor the search:

$$\sum_{c \text{ found}} \frac{|G_9'|}{|\operatorname{Stab}(\chi_c)|} \;=\; 1{,}722{,}704{,}635{,}330{,}560 \;=\; N_\chi(4,9),$$

an exact integer identity, reached exactly. Provided the enumerated classes are valid, pairwise inequivalent, and carry their true stabilizer orders, and provided the extension target is exact, a missed class leaves the sum short and it cannot overshoot; under those hypotheses the identity proves that the search visited every class, which is Theorem 2. Those hypotheses are conditions on this project's own canonicalizer and extension counter, not on the search: a defective list carrying duplicates or wrong stabilizer orders could in principle overshoot. The first three are discharged by the coverage certificate of §6, checked by a program sharing no code with the generator; the fourth is not. Of the classes found, 8,913 have a nontrivial stabilizer.

Completeness of the list is not by itself connectivity of $\underline{\mathcal{G}}^{9,4}$: those are different statements, and the second also needs the listed classes to be joined to one another. That is a separate artifact, a mutation *spanning tree* over all 9,276,595 classes (§6), recording for each class but a fixed root the parent class and the basis mutated to reach it. The same standalone checker verifies in exact arithmetic that every recorded edge is a genuine mutation edge of $\underline{\mathcal{G}}^{9,4}$ and that the parent pointers form a tree with one root, so every listed class is joined to that root by a path of mutations. All 9,276,595 of them therefore lie in one component; the list being complete, $\underline{\mathcal{G}}^{9,4}$ is connected.

The harvested holonomy is the full group: $\pi(H) = S_9$ of order 362,880 and the sign part full. Notably, on the shipped certificate the 73 stabilizer-derived generators are all even permutations and generate exactly A_9, and a single edge voltage of odd permutation part supplies the remaining coset — so on that certificate family (i) alone does not reach S_9. (We do not claim this of the complete family (i) over all classes, which was not enumerated: harvesting stops once H is full.) With $\underline{\mathcal{G}}^{9,4}$ connected, Lemma 4 gives $[\overline{G} : H] = 1$, so $\overline{\mathcal{G}}^{9,4}$, $\mathcal{G}^{9,4}$ and $\underline{\mathcal{G}}^{9,4}$ are connected, and $(9,5)$ follows by duality. This completes Theorem 1.

Remark 6. The computation was interrupted by an out-of-memory kill and resumed from a checkpoint. A resumed run harvests edge voltages only from levels expanded after the resume, so the subgroup it computes is a *lower* bound $H' \leq H$. Since $H' = \overline{G}$ was reached, $H = \overline{G}$ outright: a lower bound equal to the whole group is the whole group. The direction matters — for a negative verdict a lower bound proves nothing, and a completing pass over the earlier levels would be mandatory. The shipped summary records the lower-bound status explicitly.

6. Certificates, and what is not certified

The claims above rest on artifacts rather than on trust in the search programs, but the two halves rest on different artifacts and to different depths, and the first boundary below says how. Each completed case emits representatives, a spanning tree with voltages, generators, and exhibits; two checkers sharing no code with the generator — one pure Python, one `numpy` — re-verify the chirotope conditions, the group action, the mutation identity of every tree edge, the generator words, and the resulting group orders from scratch. Five sabotage mutations (corrupted representative, corrupted voltage, corrupted generator, truncated exhibits, re-pointed tree parent) are rejected, and the verifier fails loudly when a required artifact is missing, renamed or emptied.

What follows states the boundaries plainly, beginning with the one that is sharp enough to belong before the claims it qualifies.

The compact certificates prove $H = \overline{G}$; the catalog and its reachability are certified by a separate artifact, and only on one side of the mass identity. The compact $(9,4)$ certificates contain the root-path closure of the classes referenced by generators — a few hundred classes — which is what the holonomy claim needs and all that the checkers above verify. The left-hand side of the mass identity is a different object: the 9,276,595 classes themselves, with their stabilizer orders and with mutation edges joining them. That object is now a 10.4 MB file in the repository, and it records only the key of one root class — a 126-bit integer whose bits are the signs of a chirotope on the bases in colex order — and, for each other class, the class it was reached from together with the colex index j_c of the basis mutated to reach it.

There are no class keys in it, no stabilizer orders, no voltages and no depths: those are *derived*. What makes that possible is an identity that had been read as a check,

$$(\sigma_c, \varepsilon_c, s_c) \cdot \chi_c \;=\; \mu_{B_{j_c}}\big(\chi_{\mathrm{parent}(c)}\big) \qquad \text{for some } (\sigma_c, \varepsilon_c, s_c) \in G'.$$

It says that χ_c lies in the G'-orbit of $\mu_{B_{j_c}}(\chi_{\mathrm{parent}(c)})$; and the class key is by construction a function of the orbit alone. Read as a derivation rather than as a check, it therefore determines each class's key from its parent's. The rows are stored in a canonical order — by depth, then by parent, then by mutated basis — which is a function of the tree alone, makes every parent pointer precede its row and the parent array monotone, hence codable as a 2.3 MB bitmap, and lets the checker verify the layout itself. The earlier form of the artifact — a 62 MB array of sorted keys and stabilizer orders beside an 83 MB witness of parents, mutated bases, voltages and depths — is still what the export writes, but it is redundant, and it is not tracked.

A standalone checker `coverage_checker.py`, importing nothing from this project — it refuses to run if any module loaded into the interpreter resolves to a file beside it — and rebuilding the basis order, the Grassmann–Plücker conditions, the sign lattice and the group action from scratch, reconstructs that catalog and checks it in exact integer arithmetic. Taking the classes in order, each parent before its children, it flips the recorded basis of the parent's chirotope and verifies that the result is a *valid* uniform chirotope — which is exactly the statement that B_{j_c} is mutable in the parent, so that the edge is a real mutation edge — and then canonicalizes it into the child's key and stabilizer order, by enumerating every admissible relabelling and maximising over the sign lattice. It verifies that the root key it was handed is itself extremal in its orbit; that the 9,276,595 reconstructed keys are pairwise distinct; that every reconstructed stabilizer order divides $|G'|$ and that their histogram is the one the manifest declares; and that the orbit masses sum to 1,722,704,635,330,560 with a count of 9,276,595. On the tree it verifies that the packed streams are well formed — exact bit lengths, zero padding, one gap bit per non-root class, every basis index in range — and that each parent strictly precedes its row, so that the parent map cannot cycle and every row reaches row 0, the one parentless row, with an independent pointer-doubling pass confirming it; and that the rows are in the canonical order, so that no reordering can pass unnoticed. Finally the group element that its own canonicalization applied is read off as $(\sigma_c, \varepsilon_c, s_c)$ and handed to a second, independent implementation of the action — one built from the defining formula, with the inverse permutation and a reorientation-parity table rather than the placement machinery — which confirms the displayed identity exactly, as 126-bit sign vectors, for every one of the 9,276,594 edges.

Validity of the mutant makes B_{j_c} a mutable basis of the parent, and the child is by construction a G'-translate of that mutant, so the two classes really are adjacent in $\underline{\mathcal{G}}^{9,4}$; the identity is the witness of which translate it is, and a second implementation of the action confirms it. With the tree structure, every listed class is therefore joined to the root by a path of certified mutation edges.

This was run in full, not sampled: all 9,276,595 classes reconstructed and all 9,276,594 edges verified, in 837 seconds on two processes, all 50 checks passing — 819 of those seconds in the reconstruction, the remaining 15 in the integrity hashes, the structural checks, the root, the sort that establishes distinctness and the mass arithmetic. A reader without the legacy arrays runs the same thing minus the five cross-checks of the third qualification below. Deriving the keys rather than reading them costs nothing measurable: the 145 MB artifact, where they were given, took 853 seconds on the same machine, since it canonicalized each class only to compare against a stored key and then made a second pass for the tree. The work is sharded and checkpointed, and the rows of one depth are independent, which is what keeps it parallel although each class is reconstructed from its parent.

On a 20,000-row sub-certificate — a prefix of the canonical order, which is closed under the parent map and so is a complete certificate in its own right — twelve sabotages are each rejected: a duplicated class (an edge re-pointed onto a class already listed), a corrupted root key, a corrupted stabilizer histogram, a Grassmann–Plücker-invalid mutant, a truncated certificate, a stale hash, a parent that does not precede its row, a permutation of the parents preserving their multiset, a parent re-pointed to another class at the same depth, a malformed gap stream, an out-of-range mutated-basis index, and rows out of the canonical order. Each is built at run time by tampering with a prefix of the real certificate. Five carry a regenerated, internally consistent manifest — fresh hashes, and the count, mass and stabilizer histogram recomputed from the sabotaged data — so that only a substantive mathematical check can catch them, and the three that re-point an edge are additionally asserted at construction time to be semantically real: the tampered edge is required to become invalid or to land on a class already listed, so that a perturbation absorbed by a symmetry cannot produce a vacuous canary. A sixth corrupts nothing but the manifest's stabilizer histogram. The truncation deliberately leaves the totals unrepaired and the stale hash leaves the manifest untouched, so that the count-and-mass arithmetic and the integrity check are exercised too. The last four break the encoding, the parent order or the row order; three of them are refused outright, before any reconstruction starts, and the fourth — two rows swapped out of the canonical order — is caught there and then caught again by the mathematics downstream.

Three sabotages that the 145 MB form admitted have no analogue here, and the reason is the point of the new format: a corrupted stabilizer order, a corrupted voltage and a second parentless class are not expressible, because the certificate carries neither stabilizer orders nor voltages and its root is structurally unique. Their nearest analogues — a corrupted histogram in the manifest, a corrupted mutated basis, and a parent that does not precede its row — are in the list above.

The certificate is tracked in git, so a reader who clones the repository can check it where it lies, with no download and no regeneration; the manifest pins it by the `SHA-256` of each of its arrays and of the file itself, which is stored uncompressed with a fixed timestamp and so reproduces byte for byte. The two legacy arrays are not tracked and are not needed for any of the above. Regenerating them — about four hours for the search on four cores, six minutes for the export — reproduces the certificate exactly, and is what the cross-check of the third qualification below requires.

Three qualifications, and they are not cosmetic. First, "extremal in its own orbit" is relative to the convention recorded in the manifest, which maximises over the relabellings respecting an invariant colouring of the ground set, not over all of S_n: the unrestricted maximum differs from the reconstructed key on most classes and would cost some 160 CPU-days here. It is still a well-defined function of the G'-orbit, the colouring being invariant under reorientation and global negation and equivariant under relabelling; and that — not extremality as such — is what makes distinct keys certify distinct classes, and what makes the key of a class computable from any representative of its orbit, which is what the reconstruction uses. A reader who wants that fact rather than the word "canonical" must read the forty lines defining the colouring. Second, the checker certifies the left-hand side only. The target

1,722,704,635,330,560 enters it as a constant; an option re-adds the tracked 2,628-row extension table and confirms that it sums to that number, but nothing recomputes the extension counts, nor re-derives the $(8, 4)$ catalog they are taken over. Those are cross-checked against every published count that exists, and the identity closes exactly in the smaller cases where the answer is independently known, but they are reproducible rather than certified.

Third, and this one is new with the compact certificate: the class keys and the stabilizer orders are now *derived* by the checker, not compared against values the search recorded. That cuts both ways. A certificate cannot misreport a key or a stabilizer order that it does not contain, so a whole family of defects — and of sabotages — is no longer expressible; but the earlier arrangement had two unrelated programs agreeing, the generator's pruned canonicalization search and the checker's exhaustive enumeration, and a reader who runs only what the repository holds no longer sees that agreement. It can still be seen: the checker's `--legacy-crosscheck` compares the reconstruction against the arrays the search wrote, row by row, and on the untracked $(9, 4)$ arrays every one of the 9,276,595 rows agrees in both its key and its stabilizer order. That is corroboration of implementation agreement, not a link in the chain: nothing in the argument above appeals to it.

The consequence deserves stating exactly, because the two halves of the coverage claim now stand on different footings. That the 9,276,595 listed classes are valid, pairwise inequivalent, carry their true stabilizer orders and lie in a *single* component of $\underline{\mathcal{G}}^{9,4}$ is certificate-backed outright, under the stated canonical convention, and does not depend on the target at all: it is what the validity, distinctness, stabilizer and spanning-tree checks above establish. That they are *all* the classes is what the mass identity adds, and it is reproducible-only, since the target is. So Theorem 2, and the last step of the coverage half of Theorem 1 — from "these 9,276,595 classes form one component" to "$\underline{\mathcal{G}}^{9,4}$ is connected" — rest on a number this project computes but does not certify. Nothing else in the coverage half does.

One earlier observation was withdrawn. An intermediate report from this project described a structural phenomenon at $(9, 4)$: the sign part of the holonomy appeared to stay trivial over a large neighbourhood of the alternating oriented matroid, which was read as evidence that the labelled question there was delicate and perhaps negative. It was an artifact of a defect in the search engine, which computed the stabilizer of each newly discovered class and retained only its order, thereby harvesting family (ii) of Lemma 5 but never family (i). With the missing generators supplied the sign part is full immediately. The observation is retracted. The defect escaped notice because the smaller cases had run almost entirely through a different code path; the regression suite now forces the path that the large runs take. We record this because the failure mode is general: an incomplete harvest produces a plausible *negative* structural finding, and a negative finding is exactly what one is predisposed to believe when a counterexample has been conjectured.

Relation to [3]

Their rank-3 theorem $\overline{\mathcal{G}}^{n,3}$ is theirs and is used here; our labelled results add $(8, 4)$, $(9, 4)$ and $(9, 5)$, which is what closes $n \leq 9$. Our $\mathcal{G}$ results for $n \leq 9$ reprove theirs; the difference is not a stronger theorem but that the computation here ships certificates and checkers, whereas their connectivity computation — published in [3], with its method described there — is reported without publicly archived code, data or certificates. Their suspicion of a labelled counterexample is not refuted in general; it is refuted in the range $n \leq 9$, which is where they had verified the other two levels. Our Theorem 2 confirms the value in their Table 1 against [2]; the $(9, 3)$ entry of that table needs the correction noted in §1.

Provenance

This work was carried out by AI systems (Claude, with adversarial review by Gemini and GPT models) directing exact computation under human direction, with deliberately-failing controls in every search and independent re-verification of the serialized certificates (with the scope limits of §6). No claim here rests on unverified model output: the verdicts rest on those certificates and on the short arguments above, both checkable by the reader. The author directed the work and is responsible for the claims. The retraction in §6 was found by this process, not despite it.

References

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